Cost-Efficient Traffic Allocation for CDNs under Mixed Billing Models

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***Abstract*—Content delivery networks (CDNs) allocate time-varying traffic demand across Points of Presence (PoPs), and each assign decision affects both allocation feasibility and delivery bandwidth cost. In operational settings, this cost is shaped by mixed billing models. Some PoPs are charged by accumulated traffic volume, while others are charged by 95th percentile bandwidth after traffic is aggregated within billing cost units. Minimum commitments further create paid usage regions, making the marginal cost of assigning traffic depends on the current billing state.**

This paper studies CDN traffic allocation under mixed billing models, full demand assignment, PoP capacity, platform fanout, minimum commitments, and cost-unit aggregation. We formulate the problem and develop allocation methods that use billing state during assignments. The main method, a paid-headroom reallocation method (CAA-PHR), first builds a commit-aware allocation and then moves selected traffic into already-paid 95th percentile headroom while preserving feasibility and lowers the evaluated bill.

Using a real-world CDN dataset with 8,928 five-minute slots and 98 ISP-platform pairs, CAA-PHR reduces billing cost by 24.6% over a mode-priority baseline (MPA) and by 10.9% over commit-aware allocation (CAA). The results show that minimum commitments and cost-unit structure are most important factors in practical CDN cost optimization.

***Index Terms*—Content delivery networks, traffic allocation, bandwidth cost optimization, mixed billing models, 95th percentile billing, minimum commitment, fanout constraint.**

I. Introduction

Content delivery networks (CDNs) serve large and time-varying demand through multiple Points of Presence (PoPs). A common optimization task is to allocate traffic demand from ISP-platform pairs to available PoPs while minimizing bandwidth delivery cost. The challenge of this task comes from two parts. Operational constraints determine the feasibility of an allocation. Billing rules decide how expensive an allocation costs.

The operational side includes PoP capacity, platform fanout, and full demand assignment. Capacity limits the amount of traffic a PoP can serve in each time slot. Fanout limits the number of PoPs that a platform can actively use at the same time. This limit is useful in practice because it reduces steering overhead, monitoring load, and instability from frequent path changes. Sparse activation requirements of this form also appear in network control systems, where limiting active decisions helps scalability and stability [1]. In our setting, fanout is not a billing rule, but it changes which low-cost allocation remain feasible.

The billing side is more subtle. Contracts between the CDN operator and its network provider may combine different billing models. Some PoPs are billed by accumulated traffic volume in GB. Others may be billed by 95th percentile bandwidth in Gbps. The two modes use different measurement units, so their unit prices cannot be compared directly. Minimum commitments specify usage levels that are paid regardless of actual traffic. They are billing thresholds rather than allocation constraints. Traffic below the committed level may add no cost, while traffic above it can increase the bill. For 95th percentile billing, several PoPs may share the same cost unit, so their traffic need to be aggregated at each time slot before the billed percentile is computed.

These details make local allocation decisions hard to rank. A low listed price may not imply a low marginal cost. Avoiding 95th percentile PoPs may leave already-paid headroom unused. Filling a cheap PoP early may reduce local cost while making later demand harder to assign under capacity or fanout. At the same time, a lower bill allocation is not feasible if it is obtained by leaving traffic unserved. Thus, an effective allocator needs to track billing state during allocation while preserving feasibility.

Prior work has shown that CDN routing and allocation decisions can strongly affect delivery cost. Cost-aware request routing and bandwidth allocation have been studied under usage-based charging [2], [3]. [4] optimizes content multi-homing for both cost and performance. [5] studies centralized CDN control for real-time video delivery. Percentile billing has received particular attention because the charged value depends on the traffic time series rather than only on total volume. Online allocation under percentile charging is known to be difficult to optimize exactly [6], and recent systems develop prediction-based or online methods for practical CDN percentile billing [6], [7]. Baiji studies CDN domain planning under 95th-percentile billing, committed rates, cache-hit constraints, and locality constraints [8]. These works motivate the need to reason about percentile billing during allocation. This paper studies a complementary setting where average billing and 95th-percentile billing coexist, minimum commitments apply at different billing level, 95th-percentile PoPs are grouped by cost unit, and platform fanout should be

respected.

Our work is also related learning-assisted network optimization. Recent systems use optimization or learning to improve allocation and control at scale [9], [10]. Our learned allocator follows the same broad motivation of improving allocation decisions, but keeps the learned component limited to local ranking. Capacity checks, fanout checks, billing-state updates, and final cost evaluation remain part of the allocation system.

This paper formulates CDN traffic allocation under these mixed billing and feasibility requirements. The formulation captures full demand assignment, PoP capacity, platform fanout, average billing, 95th-percentile billing, minimum commitments, and cost-unit aggregation. We then design allocation methods that treat billing as part of the allocation decision rather than only as a final accounting step.

The main method is a Paid-Headroom Reallocation (CAA-PHR). It first builds a commit-aware allocation using estimated marginal billing cost.Then performs a paid-headroom reallocation pass. The pass identifies selected traffic on average-billed PoPs and moves it into 95th percentile billing cost units with already-paid headroom. Each move respects capacity and fanout, and the modified allocation is kept only when the evaluated bill does not increase. We also evaluate a relaxed linear-programming reference Ratio-Refined Linear Programming (RLP), a mode-priority baseline Mode-Priority Allocation (MPA), a Commit-Aware Allocation (CAA), a rank-guided learning variant CAA with Rank-Guided Allocation (CAA-RGA), and a demand-driven peak-slot heuristic Demand-Driven Peak-Slot Allocation (DD-PSA).

We design and evaluate a set of allocation models.

The paper makes the following contributions.

- We formulate CDN traffic allocation under mixed average and 95th-percentile billing, including full demand assignment, PoP capacity, platform fanout, minimum commitments, and cost-unit aggregation.
- We design allocation methods that estimate marginal billing cost under different billing models without directly comparing incompatible price units.
- We propose CAA-PHR, a paid-headroom reallocation method that reuses already-paid 95th percentile headroom after an initial commit-aware allocation.
- We evaluate the methods on a real-world CDN demand dataset with 8,928 time slots and 98 ISP-platform pairs. CAA-PHR achieves the lowest billing cost among fanout constrained methods, reducing cost by 24.6% compared with MPA and by 10.9% compared with CAA in the main experiment.

Overall, the paper shows that billing structure should be considered during CDN traffic allocation, not only a post-processing step for cost calculation. Minimum commitments and cost-unit aggregation shape marginal cost, also affect which feasible traffic assignments reduce the final CDN bill.

II. Problem Formulation

A. Problem Setup

We study the problem of allocating traffic from ISP-platform pairs to PoP nodes over discrete 5-minute time slots. Let $\mathcal{I}$ be the set of ISPs, $\mathcal{P}$ the set of platforms, $\mathcal{N}$ the set of PoPs, and $\mathcal{T}$ the set of time slots. Define $\mathcal{IP} = \mathcal{I} \times \mathcal{P}$ as the set of ISP-platform pairs. Each pair $ip \in \mathcal{IP}$ has bandwidth demand d_{ip}^t at slot $t \in \mathcal{T}$, measured in Gbps.

a) Allocation and auxiliary variables.:

$$x_{t,ip,n} \geq 0, \quad \forall t \in \mathcal{T},\ ip \in \mathcal{IP},\ n \in \mathcal{N},$$

where $x_{t,ip,n}$ is the bandwidth allocated from pair ip to PoP n at time slot t, measured in Gbps. We also introduce a nonnegative slack variable $s_{t,ip}$ to record unmet demand, means the amount of traffic from pair ip at slot t that cannot be assigned to any PoP. We use the binary variable $a_{t,p,n}$ to represent the fanout constraint where $a_{t,p,n} = 1$ indicates that platform p uses PoP n at time slot t.

b) Common constraints.: The demand of each ISP-platform pair must either be allocated or counted as unserved:

$$\sum_{n \in \mathcal{N}} x_{t,ip,n} + s_{t,ip} = d_{ip}^t, \qquad \forall t \in \mathcal{T},\ ip \in \mathcal{IP}. \quad (1)$$

PoP capacity is enforced at every slot:

$$\sum_{ip \in \mathcal{IP}} x_{t,ip,n} \leq C_n, \qquad \forall t \in \mathcal{T},\ n \in \mathcal{N}. \quad (2)$$

where C_n is the maximum instantaneous capacity of PoP n in Gbps.

Let $\phi(ip)$ denote the platform associated with demand pair ip. The binary variable $a_{t,p,n}$ is activated whenever platform p sends traffic to PoP n:

$$\sum_{ip:\phi(ip)=p} x_{t,ip,n} \leq C_n a_{t,p,n}, \quad \forall t \in \mathcal{T},\ p \in \mathcal{P},\ n \in \mathcal{N}. \quad (3)$$

and the fanout constraint is

$$\sum_{n \in \mathcal{N}} a_{t,p,n} \leq \kappa, \qquad \forall t \in \mathcal{T},\ p \in \mathcal{P}. \quad (4)$$

We set $\kappa = 6$ by default. Unserved demand is allowed in the formulation but is heavily penalized, so the allocator avoids it whenever capacity and fanout constraints permit.

B. Billing Model

PoPs are billed under either average billing or 95th-percentile billing. Let $\mathcal{N}^{\text{avg}}$ be the set of average-billed PoPs. For $n \in \mathcal{N}^{\text{avg}}$, π_n^{avg} is the price per GB and μ_n^{avg} is the minimum commitment in GB. The minimum commitment is a billing threshold paid regardless of actual usage, not an allocation constraint.

Since traffic is measured in Gbps, a time slot of length $\Delta = 300$ seconds contributes $x_{t,ip,n}\Delta/8$ GB. The average-billed volume of PoP n is

$$V_n^{\text{avg}} = \sum_{t \in \mathcal{T}} \sum_{ip \in \mathcal{IP}} x_{t,ip,n} \cdot \frac{\Delta}{8}.$$

The billed average volume is

$$r_n^{\mathrm{avg}} = \max\left(V_n^{\mathrm{avg}}, \mu_n^{\mathrm{avg}}\right).$$

For 95th-percentile billing, PoPs are grouped into billing cost units. $\mathcal{U}^{95}$ is the set of 95th-percentile cost units and $\mathcal{N}_u$ means the PoPs in unit u. Each unit has price π_u^{95} per Gbps and minimum commitment μ_u^{95} in Gbps. The slot load of cost unit u is

$$L_u^t = \sum_{n \in \mathcal{N}_u} \sum_{ip \in \mathcal{IP}} x_{t,ip,n}, \quad \forall t \in \mathcal{T}.$$

$Q_{95}(L_u)$ represents the 95th percentile of the time series $\{L_u^t : t \in \mathcal{T}\}$. The billed 95th-percentile bandwidth is

$$r_u^{95} = \max\left(Q_{95}(L_u), \mu_u^{95}\right).$$

The total evaluated cost is

$$\mathrm{Cost} = \sum_{n \in \mathcal{N}^{\mathrm{avg}}} \pi_n^{\mathrm{avg}} r_n^{\mathrm{avg}} + \sum_{u \in \mathcal{U}^{95}} \pi_u^{95} r_u^{95} + \lambda \sum_{t \in \mathcal{T}} \sum_{ip \in \mathcal{IP}} s_{t,ip}.$$

We set $\lambda = 10^7$ in the implementation, so unmet demand is used only when the capacity and fanout constraints prevent full assignment.

III. Models and System Design

This section describes the allocation methods used in the system. All methods take the same demand matrix, PoP metadata, billing parameters, and fanout limit as input. They differ in how they choose PoPs for traffic assignment. All final results are evaluated with the billing model in Section II-B.

RLP is a relaxed linear-programming reference. MPA is a simple mode-priority baseline. DD-PSA is a demand-driven peak-slot heuristic for 95th-percentile billing. CAA is the base commit-aware online allocator. CAA-RGA is a learning-guided variant of CAA. CAA-PHR is the main method, which adds a paid-headroom reallocation pass after CAA.

A. RLP: Ratio-Refined Linear Program

RLP provides an offline method. It optimizes over the full traffic demand and enforces demand balance and PoP capacity. To keep the model tractable, it does not include the binary fanout constraint. We therefore treat RLP as a relaxed benchmark rather than as a deployable policy.

For each average-billed PoP n, RLP uses a billed-volume variable y_n:

$$y_n \geq \sum_{t,ip} x_{t,ip,n} \frac{\Delta}{8}, \qquad y_n \geq \mu_n^{\mathrm{avg}}. \tag{5}$$

The first inequality converts allocated Gbps demand into GB volume, while the second applies the average minimum commitment. For each 95th-percentile cost unit u, RLP uses a billing proxy variable z_u:

$$z_u \geq \rho_u L_u^t, \quad \forall t, \qquad z_u \geq \mu_u^{95}. \tag{6}$$

Here ρ_u is a refinement ratio. The first round sets $\rho_u = 1$, so z_u upper-bounds the peak load. After the first solve, RLP computes the realized peak and 95th percentile of each cost unit and sets

$$\rho_u = Q_{95}(L_u) / \max_t L_u^t,$$

with $\rho_u = 1$ if the denominator is zero. It then solves a second round using the refined ratios. Since this refinement is heuristic, both rounds are evaluated using the exact billing model, and the cheaper allocation is kept.

Each RLP round minimizes

$$\sum_{n \in \mathcal{N}^{\mathrm{avg}}} \pi_n^{\mathrm{avg}} y_n + \sum_{u \in \mathcal{U}^{95}} \pi_u^{95} z_u.$$

Except for RLP, the methods process demand one time slot at a time. At slot t, the allocator processes platforms in decreasing aggregate demand and then processes ISP-platform pairs within each platform in decreasing demand. During the slot, it tracks residual PoP capacity and the active PoP set of each platform.

The online methods also maintain billing state. For an average-billed PoP n, B_n denotes the accumulated GB volume assigned so far. For a 95th-percentile cost unit u, L_u^t denotes the current slot load, and $\hat{h}_u$ denotes the running peak proxy used only for online scoring. The exact billing cost is recomputed after the full allocation.

B. MPA: Mode-Priority Allocator

MPA is a very simple baseline. It avoids directly comparing average-billed and 95th-percentile prices, since the units differ: average prices are per GB, while 95th-percentile prices are per Gbps. At each decision, MPA first chooses among feasible average-billed PoPs. Within that mode, it prefers lower price and then larger residual capacity. Only if no average-billed PoP is feasible does it consider 95th-percentile PoPs, using the same price and residual-capacity tie-breaks. MPA does not use minimum-commitment headroom or 95th-percentile load information.

C. DD-PSA: Demand-Driven Peak-Slot Allocator

DD-PSA is an offline heuristic for 95th-percentile billing. It first identifies high-demand slots directly from traffic demand. For each platform p, it computes the aggregate demand series D_p^t and marks the top 5% demand slots $\lfloor 0.05 \times |\mathcal{T}| \rfloor$ as predicted peak slots. During allocation, DD-PSA uses the same average-billing score as CAA. For 95th-percentile candidates, it discounts the proxy score in predicted peak slots and penalizes it outside those slots. The goal is to concentrate 95th-percentile traffic into a small set of likely peak samples instead of raising many moderate-load samples.

D. CAA: Commit-Aware Allocator

CAA chooses the feasible PoP with the lowest estimated marginal billing cost. The key observation is that traffic below a minimum commitment has already been paid for and may have zero incremental cost.

For an average-billed PoP n, suppose the allocator assigns q Gbps in the current 5-minute slot. If B_n is the cumulative volume already assigned to n in GB, the incremental cost is

$$\Delta_n^{\text{avg}}(q) = \pi_n^{\text{avg}} \left[\max\left(B_n + q\frac{\Delta}{8}, \mu_n^{\text{avg}} \right) - \max(B_n, \mu_n^{\text{avg}}) \right].$$

For 95th-percentile billing, CAA uses the running peak proxy $\hat{h}_u$ to estimate whether a new assignment raises the current billable level. For cost unit u, assigning q Gbps at slot t has proxy cost

$$\Delta_u^{95}(t, q) = \pi_u^{95} \left[\max\left(\mu_u^{95}, \hat{h}_u, L_u^t + q \right) - \max\left(\mu_u^{95}, \hat{h}_u, L_u^t \right) \right].$$

This value is zero when the assignment stays below both the minimum commitment and the current proxy peak. After each assignment, CAA updates B_n for average-billed PoPs or updates L_u^t and $\hat{h}_u$ for 95th-percentile cost units.

E. CAA-RGA: Rank-Guided Allocation

CAA-RGA studies whether learned local ranking can improve CAA's candidate ordering. It uses CAA allocations as teacher decisions. Each online decision forms a ranking group whose candidates are the feasible PoPs for the current demand item. Features include demand, residual capacity, billing mode, CAA marginal score, remaining commitment state, 95th-percentile load state, and fanout usage. We train an XGBoost ranker using the fraction of demand assigned by CAA as the relevance label.

At runtime, the learned score is used only as a small adjustment to the CAA marginal score. The allocator still enforces capacity and fanout explicitly, updates the same billing state as CAA, and is evaluated with the exact final billing model. This design keeps the learned component limited to candidate ordering rather than replacing the billing logic.

F. CAA-PHR: CAA with Paid-Headroom Reallocation

CAA-PHR is the main method. It starts from the CAA allocation and then performs a local reallocation pass. For each 95th-percentile cost unit, it first computes the evaluated billed bandwidth

$$R_u^{95} = \max(Q_{95}(L_u), \mu_u^{95}).$$

The paid headroom at slot t is

$$H_u^t = [R_u^{95} - L_u^t]_+.$$

where $[x]_+ = \max(x, 0)$. Traffic moved into this headroom does not increase the evaluated 95th-percentile cost of unit u.

The reallocation pass scans average-billed assignments with high average cost per Gbps-sample. It then tries to move selected traffic into feasible 95th-percentile PoPs with positive paid headroom. Each move respects residual PoP capacity and platform fanout. Since a local move can affect the full 95th-percentile time series, CAA-PHR recomputes the exact final cost after the pass. The modified allocation is kept only when the evaluated bill does not increase.

TABLE I
POP DATASET FIELDS.

Field	Type	Description
`pop`	string	Unique PoP identifier.
`costunit`	string	Billing group identifier.
`price`	float	Unit price. It is per GB for average billing and per Gbps for 95th-percentile billing.
`maxBw`	float	Maximum instantaneous PoP capacity in Gbps.
`mode`	string	Billing mode, either `average` or `95`.
`minComm`	float	Minimum commitment. It is measured in GB for average billing and in Gbps for 95th-percentile billing.

IV. EXPERIMENTS AND PERFORMANCE EVALUATIONS

A. Real world CDN datasets and Setup

We evaluate the allocation models using two anonymized operational datasets from a CDN provider. The first dataset, referred to as DEMAND, contains time-series traffic demand for ISP-platform pairs. Following the notation in Section II-A, each ISP-platform pair is denoted by $ip \in \mathcal{IP}$. The second dataset, referred to as POP, describes the available Points of Presence, including capacity limits, billing modes, prices, cost-unit groupings, and minimum commitments. Together, these datasets define the demand to be served and the feasible PoP resources used by the allocation algorithms.

1) Demands: The DEMAND dataset spans a 31-day billing period from March 1, 2025 to April 1, 2025, with measurements at 5-minute intervals. It contains 8,928 time slots and 98 ISP-platform pairs. Each row corresponds to one timestamp, and each column corresponds to one ISP-platform pair ip. The value in each cell is the traffic demand d_{ip}^t at that time slot, measured in Gbps.

This dataset serves as the input demand matrix for all allocation models. For each time slot, the algorithms allocate the demand of each ISP-platform pair across feasible PoPs subject to capacity and fanout constraints. The resulting allocation will be used by the billing evaluator to compute average-billing and 95th-percentile costs.

2) PoP Dataset: The POP dataset describes the PoP resources available for allocation. Each row corresponds to one PoP node. Table I summarizes the fields used in our experiments. The `mode` field specifies the billing rule of the PoP, while `costunit` identifies the billing group. For average-billed PoPs, billing is applied at the PoP level. For 95th-percentile PoPs, traffic from all PoPs in the same cost unit is aggregated at each time slot before computing the 95th percentile.

3) Experimental Setup: We compare RLP, MPA, DD-PSA, CAA, CAA-RGA, and CAA-PHR. All algorithms are executed on the same 31-day demand matrix and PoP configuration described above. The fanout limit is set to $\kappa = 6$ by default. For the heuristic methods, fanout is enforced during assignment, when no feasible PoP remains, the residual traffic

TABLE II
OVERALL ALLOCATION PERFORMANCE ON THE 31-DAY CDN DATASET. COSTS ARE SHOWN IN 10^6 UNITS.

Method	Billing Cost	Avg. Cost	95th Cost	Runtime (s)	Max Fanout	Unserved
RLP†	2.064	1.460	0.604	3040.2	10	0
MPA	2.725	2.544	0.180	17.3	6	0
DD-PSA	2.857	2.677	0.180	18.8	6	0
CAA	2.307	1.961	0.346	19.7	6	0
CAA-RGA	2.295	1.946	0.348	163.3	6	0
CAA-PHR	**2.054**	**1.708**	0.346	23.8	6	0

†RLP is an offline reference and does not enforce the fanout constraint.

is recorded as unmet demand. RLP is kept as a relaxed offline reference because it optimizes over the full demand matrix and does not enforce the fanout constraint.

All costs are computed by the common billing evaluator in Section II-B. We use billing cost as the primary cost metric, computed as the sum of average-billing and 95th-percentile charges. Unmet demand is separated from billing cost, since the penalty term is used only to indicate infeasible allocation settings and is not a real network payment. Runtime is measured as wall-clock execution time for one complete allocation over the 31-day dataset.

B. Overall Performance

Table II shows the allocation results on the 31-day CDN dataset. All heuristic methods satisfy the default fanout limit $\kappa = 6$ and serve all demand. RLP obtains a low billing cost but uses up to 10 PoPs for one platform in a time slot because the LP relaxation does not enforce fanout. Also, RLP is an offline method.

Fig. 1 shows the cost breakdown. MPA and DD-PSA keep the 95th-percentile charge low, but their average-billing cost is much higher. CAA lowers the total billing cost by using minimum-commitment information during allocation. CAA-PHR gives the lowest cost among the fanout-constrained methods. Compared with CAA, it mainly reduces the average-billing component while leaving the 95th-percentile cost almost unchanged. This suggests that the reallocation pass moves part of the traffic from expensive average-billed PoPs into paid 95th-percentile headroom.

CAA-RGA reduces cost compared with MPA and DD-PSA, but it does not improve over CAA in this experiment. The learned ranker changes the local candidate ordering learned from CAA decisions, but it does not directly optimize the final billing objective. Overall, CAA-PHR gives the best cost among the methods that satisfy the fanout limit, with a small runtime increase over CAA.

C. Ablation of Paid-Headroom Reallocation

CAA-PHR only considers a selected fraction of expensive average-billed assignments during the reallocation pass. We vary this fraction, denoted by `top_ratio`, to study how the scope of reallocation affects cost and runtime. A value of zero corresponds to CAA without paid-headroom reallocation.

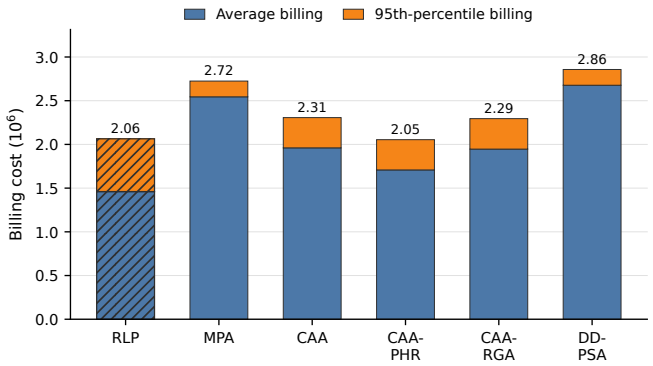


Fig. 1. Billing-cost breakdown of each allocation method. The bars separate average-billing and 95th-percentile charges. Hatched bars indicate the offline RLP method.

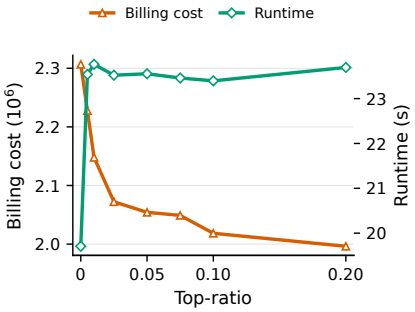


Fig. 2. Effect of the CAA-PHR reallocation scope. A larger `top_ratio` allows more high-cost average-billed assignments to be considered for paid-headroom reallocation.

Fig. 2 shows that increasing the reallocation scope reduces billing cost, while runtime changes only mildly. The reduction comes from moving more average-billed traffic into already-paid 95th-percentile headroom. The 95th-percentile charge remains nearly unchanged, indicating that CAA-PHR mainly reduces average-billing cost rather than shifting cost to the 95th-percentile side. In the main experiment, we use `top_ratio`=0.05 as a conservative default.

D. Sensitivity and Robustness Analysis

We further examine how the allocators behave when the billing contract, traffic demand, and fanout constraint change. For billing-cost comparisons, we include only operating points where all methods serve the full demand. When tighter settings that create infeasibility, we examine unmet demand separately.

Fig. 3(a) varies the minimum-commitment level. CAA-PHR gives the lowest cost over most settings. The gain is largest when committed capacity creates useful paid headroom. When the commitment multiplier becomes very high, the commit-aware methods converge because a larger part of the bill is fixed by the contract rather than by allocation decisions.

Fig. 3(b) scales the traffic demand. Within the feasible range, CAA-PHR remains the lowest-cost method. Its advan-

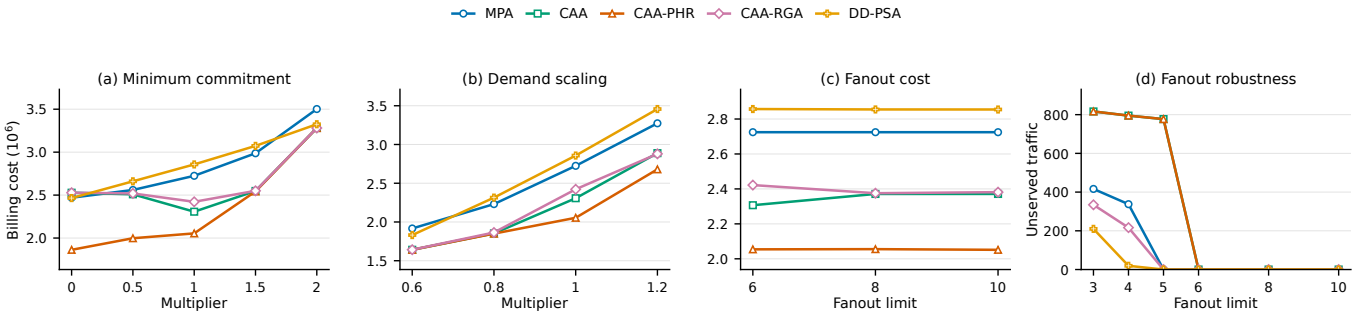


Fig. 3. Sensitivity and robustness analysis. Panels (a)–(c) show billing cost under feasible operating points. Panel (d) shows unmet demand under different fanout limits.

tage comes from reducing average-billing cost while keeping the 95th-percentile charge under control. This trend suggests that paid-headroom reallocation remains useful as demand grows, as long as the system still has enough capacity and fanout flexibility to serve all traffic.

Fig. 3(c) studies the fanout limit in the feasible region. Once the fanout limit reaches the default value $\kappa = 6$, increasing fanout further gives only limited additional cost reduction. This indicates that most of the useful allocation flexibility is already available at the default fanout level.

Fig. 3(d) focuses on feasibility under tight fanout limits. Small fanout limit lead to unmet demand, but the amount of unmet demand differs across allocators. This shows that local cost minimization and feasibility robustness are not the same objective. A cost-aware greedy decision can occupy fanout or capacity in a way that prevents later demand from being placed. CAA-RGA reduces unmet demand under some tight-fanout settings, while CAA-PHR gives the best billing cost once the setting becomes feasible.

V. Conclusion

This paper studied traffic allocation in operational CDN systems under mixed billing contracts. The results show that CDN allocation should not be viewed only as a capacity balancing problem. Average billing, 95th-percentile billing, minimum commitments, cost-unit aggregation, and platform fanout jointly shape the cost of an allocation.

We formulated the allocation problem with these billing and feasibility constraints, and designed a set of allocation models that expose different tradeoffs. We proposed CAA-PHR, which starts from a commit-aware allocation and then reuses already-paid 95th-percentile headroom through local reallocation.

Experiments on a real-world CDN demand dataset show that commit-aware allocation can substantially reduce billing cost. CAA-PHR achieves the lowest cost among fanout-constrained methods by reducing average-billing cost while keeping the 95th-percentile charge stable. The sensitivity study further shows that minimum commitment and paid headroom are central to CDN cost structure. It also shows that billing cost must be interpreted together with feasibility, since tight fanout or capacity constraints can leave demand unserved.

Future work will study feasibility-aware lookahead and cross-period learning so that allocation policies can better generalize across changing traffic and contract conditions.

References

[1] O. G. Aliu, A. Imran, M. A. Imran, and B. Evans, "A survey of self organisation in future cellular networks," *IEEE Communications Surveys Tutorials*, vol. 15, no. 1, pp. 336–361, 2013.

[2] V. Khare and B. Zhang, "CDN Request Routing to reduce network access cost," in *37th Annual IEEE Conference on Local Computer Networks*. Clearwater Beach, FL, USA: IEEE, Oct. 2012, pp. 610–617.

[3] M. Adler, R. K. Sitaraman, and H. Venkataramani, "Algorithms for optimizing the bandwidth cost of content delivery," *Computer Networks*, vol. 55, no. 18, pp. 4007–4020, Dec. 2011.

[4] H. H. Liu, Y. Wang, Y. R. Yang, H. Wang, and C. Tian, "Optimizing cost and performance for content multihoming," in *Proceedings of the ACM SIGCOMM 2012 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communication*. Helsinki Finland: ACM, Aug. 2012, pp. 371–382.

[5] M. K. Mukerjee, D. Naylor, J. Jiang, D. Han, S. Seshan, and H. Zhang, "Practical, Real-time Centralized Control for CDN-based Live Video Delivery," in *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication*. London United Kingdom: ACM, Aug. 2015, pp. 311–324.

[6] H. Chen, H. Zhan, H. Tan, H. Xu, W. Shan, S. Chen, and X.-Y. Li, "Online Traffic Allocation Based on Percentile Charging for Practical CDNs," in *2022 IEEE/ACM 30th International Symposium on Quality of Service (IWQoS)*, Jun. 2022, pp. 1–10.

[7] H. Zhan, H. Tan, H. Ni, H. Xu, W. Shan, and X.-Y. Li, "Traffic Allocation for Percentile Charging in CDNs," *IEEE Transactions on Networking*, vol. 34, pp. 857–871, 2026.

[8] J. Vanerio, H. Liu, Q. Zhang, and S. Schmid, "Baiji: Domain Planning for CDNs under the 95th Percentile Billing Model," in *2024 IFIP Networking Conference (IFIP Networking)*. Thessaloniki, Greece: IEEE, Jun. 2024, pp. 1–9.

[9] Z. Xu, F. Y. Yan, R. Singh, J. T. Chiu, A. M. Rush, and M. Yu, "Teal: Learning-accelerated optimization of wan traffic engineering," in *Proceedings of the ACM SIGCOMM 2023 Conference*, ser. ACM SIGCOMM '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 378–393. [Online]. Available: https://doi.org/10.1145/3603269.3604857

[10] Z. Yang, G. Fan, A. Cao, Y. Guo, W. Li, T. Liu, and H. Jin, "Learning to Accelerate Traffic Allocation Over Large-Scale Networks," in *IEEE INFOCOM 2025 - IEEE Conference on Computer Communications*, May 2025, pp. 1–10.